Implementation of Agile Methodology based on SCRUM Tool

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***Abstract*— The enlargement for developing of software project in the IT Companies by habituated with traditional waterfall methodology, a software developers are facing much problem related to the software requirements. So with respect to that problems the software developers/teams deal with an Agile methodology to facilitate fully satisfaction of the customer related to the software requirements of the projects because of dynamic changing. Therefore in this paper we gathered the data related to the traditional waterfall as well as newly Agile methodology for success, challenged and failed rates from various blogs, laboratory and IT companies from 2010 to 2016 shown in Table1 and calculated these two methodologies rates through percentage to show how Agile methodology is working excellent for developing the software project that is the success, challenged and failed rates of methodologies are in agile (54%, 37%, 9%) whereas in waterfall has (35%, 45%, 20%) which is presented in Graph 1 and Graph 2. Hence this research has gathered all the 19 possible features/methods which should be taken care at the period of software development time, that is mentioned in Table 2 associated with their possible values that are *Low/Less/No =1, Medium/Intermediate =2, High/Yes =3* and we evaluated its values to find betterment of software quality, so in traditional waterfall methodology has only 61.40% but in Agile methodology has 82.45% which is giving better performance in IT industry. Therefore we taken a dataset related to course registration system and implemented with an SCRUM tool that is axosoft tool which describes all the sprint of the project related to the iteration to handover the work of each developer for complete their task on time.**

Keywords*— *Agile Methodology, Iterative, Incremental, Success rate, Challenged rate, Failed rate, SCRUM, Axosoft tool

I. INTRODUCTION

Agile software methodology is mainly alternate solution of waterfall methodology where traditional methodologies [6] face difficulties at the time of software development. The main objective of Agile methodology is to develop better project in shorter period of time because it follows the iterative along with incremental process [3, 6] to develop the project and also maintain the time-frame [9] and only in given time-frame a developer has responsibility to complete the project with their proper functionalities to deliver the customer. The agile methodology main target is to involve the customer more and more so that at the time of software development [5] a customer understand and analyze the software requirements and also if developer faces any difficulties regarding functionalities then customer and developer discuss with each other to make the project more improvable by implementing more iteration that's why Agile methodology has no need to maintain fully documentation to hand over at project deliver time because a clients/customers are already familiar with their project functionalities by involving [1] them at development phase with high communication [8] due to which it has minimal documentation.

II. PRESENTATION ANALYSIS OF SUCCESS, CHALLENGED AND FAILED RATES

Here we studied about project success, challenged and failed rates [5] of agile and waterfall methodologies to review from various laboratory, blogs and IT industries which are given.

- **Successful:** Deliver successfully working project having all required features and functions on time with given budget.
- **Challenged:** Deliver project late in compare of specified time and out of budget, also without working all functions/ features properly.
- **Failed:** The product deliver that are not usable for customer so it's about to canceled.

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III. PRESENTATION RATES OF METHODOLOGY AS LITERATURE SURVEY

Organizations	Years	Agile Methodology			Waterfall Methodology		
		Success	Challenged	Failed	Success	Challenged	Failed
Dr. Bobb's, The World of Software Development[41]	**2010**	60%	28%	12%	47%	36%	17%
InfoQueue [42]	**2011**	67%	27%	6%	50%	36%	14%
Zen Ex Machina Pty Ltd[43]	**2012**	43%	45%	12%	26%	59%	15%
Ambysoft's Project Success Rates Survey[45]	**2013**	64%	28%	8%	49%	33%	18%
Non-Existent Software Crisis: Debunking the Chaos Report [46]	**2014**	64%	30%	6%	49%	32%	18%
The CHAOS report from the Standish Group[47]	**2015**	42%	49%	9%	14%	57%	29%
VITALITY Chicago Inc. [48]	**2016**	39%	52%	9%	11%	60%	29%
Total Percentages=		**54%**	**37%**	**9%**	**35%**	**45%**	**20%**

Table 1: Calculating Success, Challenged and Failed Rates

Now we reached at actual value to calculate the success, challenged and failed rates of these two Agile and Waterfall Methodologies. So that we gathered the last seven years record from various software development laboratories, blogs and IT industry. Here first we find the average of these percentage of project to detect the overall percentage of the project performance.

The average of **Success Rate** of Agile Method= (60+67+43+64+64+42+39)/7 =379/7 =54.14=54

The average of **Challenged Rate** of Agile Method= (28+27+45+28+30+49+52)/7=259/7=37

The average of **Failed Rate** of Agile Method= (12+6+12+8+6+9+9)/7 =62/7 =8.85 =9

So, the average rate of all **success, challenged, failed rate of Agile methodology** that comes after calculating all these rates (54, 37, 9).

The average of **Success Rate** of Waterfall Method= (47+50+26+49+49+14+11)/7 =246/7 =35.14 =35

The average of **Challenged Rate** of Waterfall Method= (36+36+59+33+32+57+60)/7 =313/7 =44.71 =45

The average of **Failed Rate** of Waterfall Method= (17+14+15+18+18+29+29)/7= 140/7 =20

So, the average rate of all **success, challenged, failed rate of Waterfall** methodology that comes after calculating all these rates= (35, 45, 20).

Hence overall **success, challenged, failed rates** of **Agile** and **Waterfall** methodologies of last seven years from 2010 to 2016 are calculated hare that is **(54, 37, 9)** and **(35, 45, 20)**. So here we found that in every prospective Agile methodology is better than waterfall methodology because Agile has more success rate than waterfall, less challenged rate than waterfall and less failure rate in agile that is only 9% but in waterfall it reached to 20%. So by calculating of these values we reached to conclusion that is Agile methodology is better than waterfall methodology.

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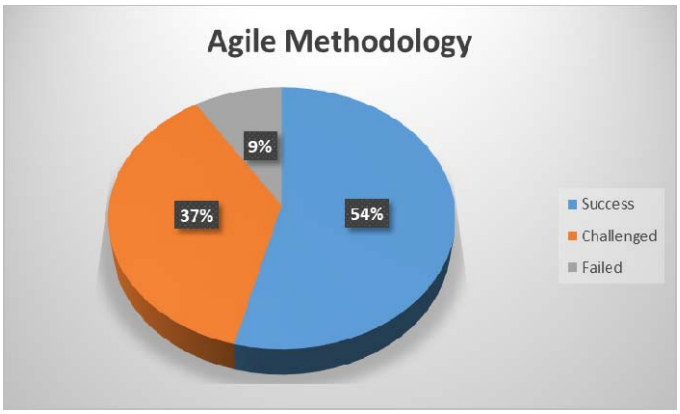
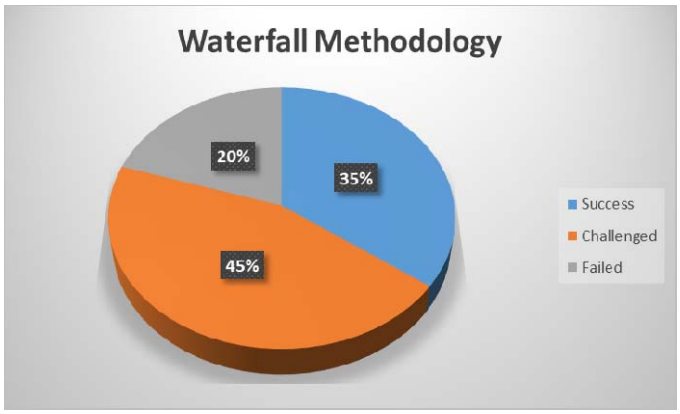


Graph 1: Graph for Success, Challenged and Failed Rates of agile methodology



Graph 2: Graph for Success, Challenged and Failed Rates of waterfall methodology

IV. Results analysis of Agile and Waterfall Models

Note: *Low/Less/No =1, Medium/Intermediate =2, High/Yes =3*

Features/Methods	Waterfall Model	Values	Agile model	Values
Requirement Specification	Beginning (High)	3	Frequently Changed (Medium)	2
Understanding of Requirements	Well Understand (Medium)	2	Not Well understood (Low)	1
Cost of the Projects	Project starts from beginning if changes (High)	3	Re-write the program (Intermediate)	2
Availability of reusable component	No	1	Yes	3
Complexity of system	Simple	1	Simple	1
Risk Analysis	Only at beginning (Low)	1	No Risk Analysis (High)	3
Users Involvement in all phases of SDLC	Only at beginning (Low)	1	All the Iterations (High)	3
Guarantee of Success	Intermediate	2	High	3
Overlapping Phases	No overlapping	1	Overlapping (yes)	3
Implementation time	Long (High)	3	Intermediate	2
Product Flexibility to change need	Low	1	High	3
Changes Incorporated	Difficult (No)	1	Easy (Yes)	3
Expertise Required	High	3	High	3
Level of trust between team members	More focus on processes and tools (Low)	1	More focus individuals and interaction (High)	3
Cost Control	Yes	3	Intermediate	2
Product Quality	Low	1	High	3
Customer Satisfaction	Low	1	High	3
Documentation	Sufficient (High)	3	Minimal (Low)	1
Resource Control	Yes	3	Yes	3
Total value over feature/method of models=		**35**	**Total=**	**47**
Percentage of these values over its features/methods	**Percentage of Waterfall Model**	**61.40%**	**Percentage of Agile Model**	**82.45%**

Table 2: Calculating Agile and Waterfall methodologies performances

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Here, there are 19 features/methods whose values are given with their ***Low/Less/No=1, Medium/ Intermediate =2, High/Yes =3*** criteria. So total value for all the feature/methods is 19*3=57 to calculate the percentage of Waterfall and Agile methodologies.

The percentage of **Waterfall Model** = (35*100)/57 =61.40%

The percentage of **Agile Model** = (47*100)/57 =82.45%

Methodologies	Agile Model	Waterfall Model
Percentage of methodology over performances	82.45%	61.40%

Table 3: Final Result of agile and waterfall methodologies

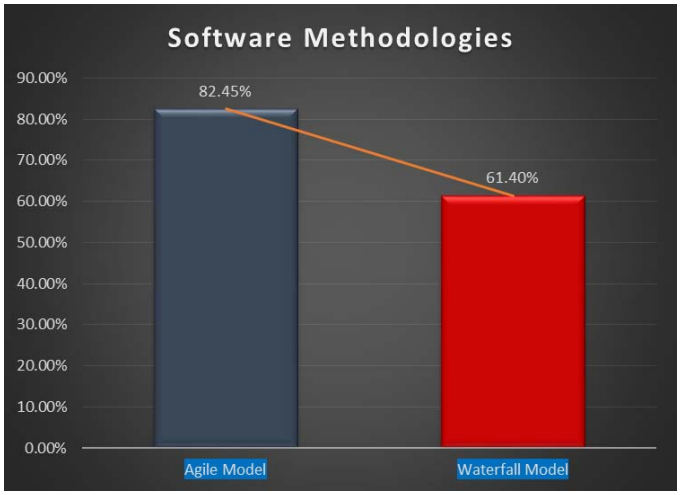

Chart 1: Comparison Chart of Agile and Waterfall methodologies

Here to calculate whole the features/methods of agile and waterfall methodologies we taken support of various properties that includes their requirements, cost, flexibility, overlapping phases, performances, customer involvement, customer satisfactions level, product quality and also related to documentation. Therefore after calculating of all these values with their performances we reached to the conclusion that is a major differences between agile and waterfall methodologies founded, so agile gives more performances over waterfall methodology. So these performances are making more excellence for IT industries by implementing Agile methodologies because it has 82.45% but in waterfall methodology has only 61.40% which make worst for developer and customer.

V. SCRUM

SCRUM introduced as a framework initiated by Ken Schwaber [1, 36] that primarily focuses to develop complex systems [3] and products based on empirical process as well as control theory, the term empirical process means inspect and adapt that is do something, check and adapt then again do something else better, check and adapt to improvise the product. SCRUM is grounded on the basis of iterative and incremental process [7] to improve the predictability, reduces the error and control the risk which occur during software development time.

There are three main roles in SCRUM [36] that includes ***product owner***, ***scrum master*** and ***team member or scrum team***. So ***product owner*** are only responsible to maintain all the list of features/requirements [12] that are gathered by end user, team and stakeholder and come up with the product backlog highly prioritized [6] by product owner [7] where product owner is only authenticated person to change or update any product features after completion of each iteration. ***Scrum teams*** are typically consist 7+/-2 people means maintain either 5 or 9 people to work on project, here team maintains the cross functional operation to include designer, coder, tester and etc. and in each particular iteration team completes all life cycle to complete the task. So the team member will make commitment [7] to complete their work on given time period and at the end of each day by arranging of a daily basis meeting [12] they discuss what they have done today and what they will do tomorrow for project development. ***Scrum master*** treats as a servant leader of the team to protect them from external or internal disturbance like if product owner keeping interest to add a new feature at the mid of the sprint then scrum master protects them to wait until completion of the current iteration [36]. Scrum master are highly responsible to train the product owner and scrum team to follow the all principles of agile methodology.

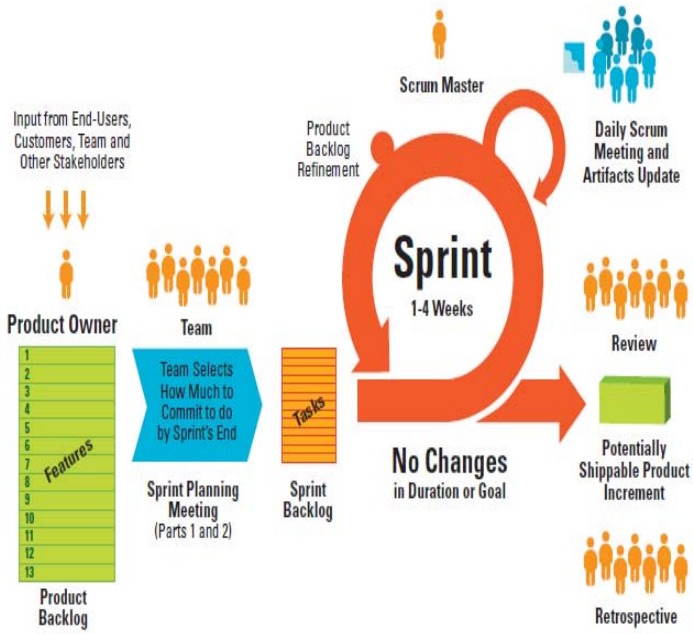

Figure 1: SCRUM process overview

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VI. AXOSOFT TOOL

Axosoft is determined as an OnTime software to manage agile software project that is highly popular tool to assure the customer to complete their project on specific time that includes the feedback given from customer and fix the issues of user to minimize the bug. Axosoft tool facilitates the user to assemble all the requirements of the project and with the help of backlogs work with all the sprint to burndown the chart, here teams are assigned with their fixed time to complete its work. It creates the user story prioritized by product owner having time-frame to complete the task.

VII. DATASET: COURSE REGISTRATION SYSTEM

Here a dataset have been taken related to course registration system for the student having their various requirements that are related to the student to be used at registration time for the sake of enroll the student in any courses. The requirements are *student test case*, *instructor* and *department head* for inserting, viewing and dropping the courses so that are given below.

Test Case No.	Requirements
	Student Test Cases
TC1	Register
TC2	Add Course
TC3	Drop Course
TC4	View Course
TC5	Edit Course
TC6	View Grade of semester
	Instructor
TC7	View Current Section Registration
TC8	View Enrolled Students
TC9	Assign Grade
TC10	View Grading section
TC11	Decrease/Increase Grade
	Department Head
TC12	View Catalog as Dept. Head
TC13	Add New Courses as request
TC14	Add request that does not exist
TC15	Delete Requests

Table 4: Data Set for Course Registration System

Here an agile SCRUM tool that is axosoft is used to implement all the requirements which are listed in course registration system dataset in Table 4.

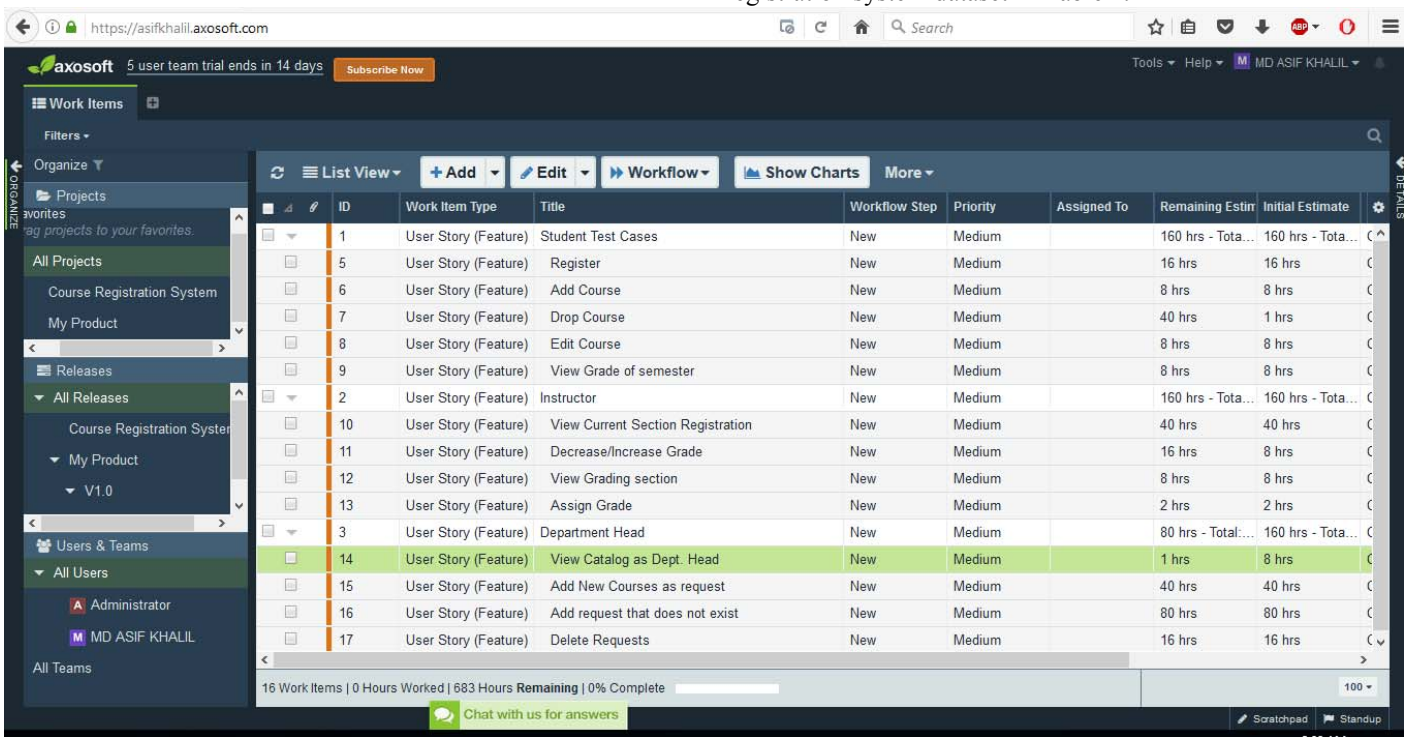

Figure2: Dataset implemented on axosoft SCRUM tool

VIII. CONCLUSION AND FUTURE SCOPE

Having tremendous growth of IT project development, a project developers are really facing many problems so this paper includes the survey of success, challenged and failed rates of traditional waterfall methodology to compare with newly agile methodology that highly working in IT industry, so here we calculated the success, challenged and failed rates of these two methodologies through which Agile methodology is obtained as best methodology to use for software development. Due to which our research includes all the possible features/methods in Table 2 that are taken here with their observed value from various journal's paper to judge how Agile fit in IT industry by calculating both methodologies features/methods and displayed in Table 3. Therefore a dataset

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of course registration system has been taken to fit all the user stories of the software with the help of SCRUM related axosoft tool to divide all the requirements associated with its time to understand the team that how much expected time a project will take and also assure the customer to complete the project on time.

Having habituated with Agile Methodology highly communication is required if communication of developer and customer is not well understand then Agile is not recommended for software development in the IT industry.

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